

Karthik Arun

iamkarthikarun.github.io +91-79040 56987 | karthikarun2000@gmail.com | linkedin.com/in/karthikarunr |

Data Scientist and AI Engineer building autonomous, LLM-powered systems for enterprise and aviation environments, from an aircraft-defect chatbot at Singapore Airlines to an autonomous CRM agent at Walmart Global Tech, with a focus on reliability and measurable business impact.

EXPERIENCE

Software Development Engineer III - ML/AI

Jul 2025 – Present

Walmart Global Tech

Bangalore, KA, India

- Engineered a 4-class seller segmentation model using a robust ensemble architecture, resolving a highly volatile top-tier segment to achieve a 75% precision-recall threshold
- Architected a two-stage NLP entity resolution pipeline for fuzzy Name/Address matching, serving models via Vertex AI and processing at scale on DataProc-managed Spark with BigQuery as the source/sink; utilized diverse edit distance metrics to cross-reference Walmart sellers with external systems, driving an 18-percentage-point lift over baseline accuracy
- Shipped the debut release of BD-Copilot, an autonomous agent that executes Salesforce workflows on-demand, doubles as a documentation-search chatbot advisor, and reports health through a Grafana/Prometheus/OpenObserve telemetry stack; now developing its next iteration, a dynamic UI that adapts in real time to user preferences.

Associate Data Scientist

Apr 2025 – Jul 2025

Daimler India Commercial Vehicles

Chennai, TN, India

- Built hierarchical forecasting pipeline for 1,800+ warranty claim series using Nixtla and Spark-distributed deep learning and statistical models with cross-validation for anomaly-resistant predictions
- Engineered production-scale failure rate analytics pipeline processing 500K+ warranty claims using Spark, reducing analysis time from days to hours with interactive Nevada plot visualizations
- Discovered part failure relationships using FP-Growth association mining and Louvain clustering, identifying 50+ high-risk part communities for predictive maintenance optimization

Data Science Intern

Mar 2024 – Aug 2024

Singapore Airlines

Singapore

- Built a hybrid-inference aircraft-defect chatbot combining AWS Bedrock with self-hosted, quantized LLMs for low-latency, data-sovereign responses, cutting time-to-fix by 30%
- Engineered a retrieval-augmented generation pipeline with dense-sparse hybrid search, lifting document-retrieval accuracy by 40%
- Replaced AWS Textract with Claude-3 Haiku for OCR in the document processing pipeline, cutting monthly costs from \$12 to \$0.05 per 1,000 documents while maintaining 95% accuracy

Salesforce Analyst

Jul 2021 – Jul 2023

Deloitte Consulting

Chennai, TN, India

- Developed and deployed large-scale predictive models using AWS SageMaker for customer churn prediction, identifying 85% of at-risk customers 2 months in advance
- Built recommendation systems for vehicle service packages using Salesforce Einstein AI, increasing cross-sell revenue by \$2.5M annually
- Engineered scalable model serving pipelines using AWS Glue, processing 2TB+ of customer data with 40% reduced latency
- Built a CI/CD system from the ground up for cloud deployments, using BitBucket Pipelines, SFDX, Jenkins and Docker, improving deployment times by 97%

TECHNICAL SKILLS

Generative AI and Agentic Systems: AWS Bedrock, Vertex AI, LangChain, LangGraph, HuggingFace

Machine Learning: Python, SQL, PyTorch, Scikit-learn, Nixtla

Cloud and Infrastructure: AWS (SageMaker, Glue), GCP (DataProc, BigQuery)

Engineering and Deployment: Docker, FastAPI, Jenkins, Airflow, Git, Shell Scripting

Monitoring and Observability: Grafana, Prometheus, OpenObserve, MLflow

Data and Visualization: MySQL, PostgreSQL, MongoDB, Neo4j, Tableau, Plotly

CERTIFICATIONS

AWS Certified Machine Learning Associate <i>Amazon Web Services</i>	Oct 2024
Salesforce Certified AI Associate <i>Salesforce</i>	Oct 2024

EDUCATION

National University of Singapore <i>Master of Technology in Enterprise Business Analytics</i> Cumulative GPA: 4.75/5.0; IBM Medal for Best Outgoing Student	Singapore <i>Aug 2023 – Oct 2024</i>
SASTRA University <i>Bachelor of Technology - Information and Communication Technology</i>	Thanjavur, TN, India <i>Jul 2017 – Jul 2021</i>

PROJECTS

TurboPredict <i>Singapore Airlines R&D</i> <i>PyTorch, Docker, Airflow, FastAPI</i> Built an end-to-end ML pipeline for real-time turbofan engine health monitoring, implementing automated model retraining and deployment pipelines	May 2024
QuizMasterAI <i>NUS-ISS</i> <i>LangChain, Transformers, Qdrant</i> Fine-tuned an open-source LLM for question generation and evaluation in a school setting, using PEFT techniques and implementing efficient serving architecture and engineered prompts	Mar 2024
GraphRecommend <i>NUS Co-op Store</i> <i>PyTorch Geometric, Neo4j, FastAPI</i> Developed a production-ready recommender system using Graph Neural Networks, serving 500k+ users with sub 50ms latency	Feb 2024